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Analysis of keyhole dynamics and laser energy absorption in Ti6Al4V deep penetration laser welding process: An integrated numerical simulation and statistical approach

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Highlights

- A methodology integrating CFD simulation with KDE is developed.
- An analysis method based on the peak probability density is established.
- The probabilistic nature of keyhole morphology and laser absorption is characterized.
- The mechanism of synchronous fluctuations in keyhole and laser absorption is revealed.

Outline

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Abstract

Deep penetration laser welding (DPLW) process exhibits nonlinear keyhole dynamics and dynamic energy absorption characteristics, significantly influencing the weld quality and defect formation. In this work, an integrated framework combining numerical simulation and statistical approach was presented to analyze the keyhole behavior and laser energy absorption. A high-fidelity multi-physics model was developed to capture the interaction between the laser beam and the material using the improved ray tracing method. Kernel density estimation (KDE) was implemented for probabilistic characterization of simulation data of keyhole dynamics and laser energy absorption. Additionally, a perspective is proposed for understanding keyhole dynamics by considering the point of the peak probability density as the equilibrium center (a relatively stable state) of stochastic fluctuation behavior. Statistical analysis revealed that the keyhole dynamic parameters and laser energy absorption in the fluctuation stage underwent stochastic fluctuation behavior around the equilibrium point corresponding to the maximum probability density. The statistical analysis provides insights into the probabilistic nature of the keyhole behavior and laser energy absorption. Additionally, the synchronous variations in the keyhole behavior and laser energy revealed the underlying mechanisms of the fluctuations around the equilibrium state. This work develops an integrated framework of simulation and statistical analysis to study keyhole dynamics and laser energy absorption, providing a new perspective for understanding the physical mechanisms underlying the DPLW process.

Keywords

Numerical simulation; Deep penetration laser welding; Laser energy absorption; Keyhole dynamics; Statistical analysis

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CRediT authorship contribution statement

Declaration of competing interest

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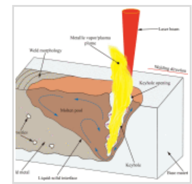

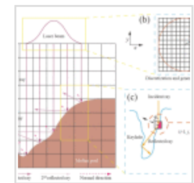

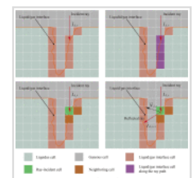

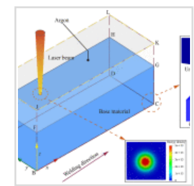

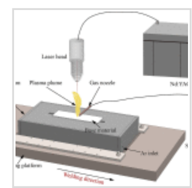

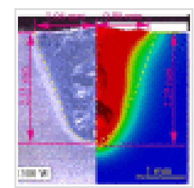

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1. Introduction

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In deep penetration laser welding (DPLW) process, the laser-induced keyhole forms due to the rapid evaporation of the molten metal under direct laser radiation [1,2]. The keyholing phenomenon, an essential feature of the DPLW process, can induce strong fluctuations at the liquid-gas interface, thereby dramatically increasing the intricacy of laser welding system [[3], [4], [5]]. The keyhole instability can lead to welding defects and consequently degrade the welding quality. In the laser welding and additive manufacturing (AM) community, keyhole dynamics are intrinsically associated with the transient laser energy absorption coupled to laser welding system [[6], [7], [8]]. Despite such significance, the interrelation mechanism between the laser energy absorption and keyhole dynamics has not been fully understood. A complete description of keyhole dynamics and laser energy absorption is critical to the welding quality and laser welding process monitoring.

Extensive works have been performed to reveal the complex laser-matter interaction by utilizing experimental methods in various laser processing processes, including laser welding and additive manufacturing (AM) processes such as laser powder bed fusion (LPBF). Optical imaging technology has been prevalently utilized to capture the transient keyhole behaviors through the coaxial or lateral configuration monitoring methods [[9], [10], [11], [12]]. These investigations basically clarify the issues of the keyhole fluctuations [9,10,13], keyhole instability [9], and interrelation mechanism between the keyhole and metallic vapor/plasma plume [11,13]. However, due to the inherent opacity of the subsurface phenomena, traditional viable monitoring methods encounter significant experimental challenges in directly observing the transient keyhole behaviors without disturbing the manufacturing process [14]. This challenge is particularly pronounced in LPBF process where the powder bed environment and complex interaction further complicate direct observation. As an emerging method for in-process probing subsurface phenomena, high-speed X-ray imaging has facilitated the understanding of keyhole dynamics and keyhole porosity formation mechanism in laser welding [15] and LPBF processes [16,17] at an unprecedented scale. Additionally, optical coherence tomography

(OCT) has been successfully utilized to reconstruct 3D morphology of the keyhole [9,18]. The complementary information of subsurface phenomena revealed by OCT and high-speed X-ray imaging technique has facilitated further studies of keyhole dynamics in laser welding and LPBF processes. However, due to the unmeasurable nature of laser coupling within the highly dynamic keyhole, there are difficulties in exploring the connection between transient keyhole behaviors and dynamic laser absorption through experimental methods. Recently, the causal correlation between instantaneous keyhole morphology and dynamic laser absorption has been explored by utilizing the combined application of in-situ absorption measurement such as calorimetry [19] and integrating sphere methods, along with imaging techniques in laser welding and LPBF processes [[20], [21], [22]]. The highly dynamic keyhole was found to be strongly coupled with laser energy absorption under different laser processing conditions and metal materials. Consequently, it is of significance to investigate keyhole dynamics, the energy coupling issue, and intrinsic relationship between them in the broader context of laser-based manufacturing processes. However, quantitative insights into dynamic energy coupling remain limited due to the experimental challenges in measuring dynamic optical absorptance [23].

Numerical simulations have emerged as a potent tool for insight investigations of keyhole dynamics and dynamic energy coupling. Extensive mathematical models have been developed to address various challenges associated with the complex underlying physics in laser-based manufacturing processes, particularly the transient dynamics of the laser-induced keyhole [24,25]. Recently, some numerical models considering the energy coupling have been developed to reveal dynamic laser energy absorption [8,22,26]. Additionally, laser energy absorption was demonstrated to be linked to the keyhole porosity generation [27]. Further studies have demonstrated that the spatial distribution of laser absorption was tightly linked to the keyhole morphology and its fluctuations [[28], [29], [30]]. In conclusion, visualization-based works including in-situ monitoring and multi-physics models have offered powerful tools for insight investigations of keyhole dynamics. However, critical information regarding the keyhole dynamics, laser energy absorption, and interrelation mechanism between them needs further investigations. Furthermore, existing researches mainly focus on the short-term or dynamic changes of the system in the evolution process to analyze the interaction between the laser beam and material. However, this approach might not fully capture the long-term statistical patterns that could emerge from the randomness involved. Some studies have analyzed the random and time-varying characteristics of the laser keyhole dynamics and laser absorption from the perspective of statistical analysis [31]. Further investigation is needed to apply statistical methods to analyze the regularities within the random events of the dynamic behaviors and laser energy absorption for the keyhole.

In this work, high-fidelity multi-physics simulations were conducted to model the laser-material interaction in Ti6Al4V DPLW process using the improved ray tracing method. A novel integrated framework combining high-fidelity multi-physics simulations with KDE-based statistical analysis was developed to characterize the temporal evolution and probabilistic distributions of the keyhole morphology and laser energy absorption. The fluctuating behaviors of the keyhole and laser absorption around their respective equilibrium states were revealed from the probabilistic perspective. Furthermore, the underlying physical mechanisms governing the fluctuations around the equilibrium states were elucidated by the synchronous analysis of the keyhole behavior and laser absorption. The proposed integrated framework enables a deeper understanding of the probabilistic nature of the keyhole fluctuations and laser energy absorption in the DPLW process.

2. Methodology

As illustrated in Fig. 1, many coupled physical phenomena including melting, evaporation, ionization, and solidification are involved in the DPLW process, which can affect heat and mass transfer behavior, as well as fluid flow dynamics. Consequently, a high-fidelity multi-physics model is developed to model the keyhole behavior and laser-material interaction in the DPLW process. To simplify the model, the following assumptions are adopted: (1) the molten metal within the molten pool is treated as incompressible Newtonian fluid exhibiting laminar flow behavior. (2) the buoyancy effect is addressed using the Boussinesq approximation. (3) the mushy zone method is implemented to account for the influence of partial solidification on the fluid flow during the solidification transition. This section sequentially outlines the governing equations, laser heat source modeling, simulation setup, and material properties for the developed model.

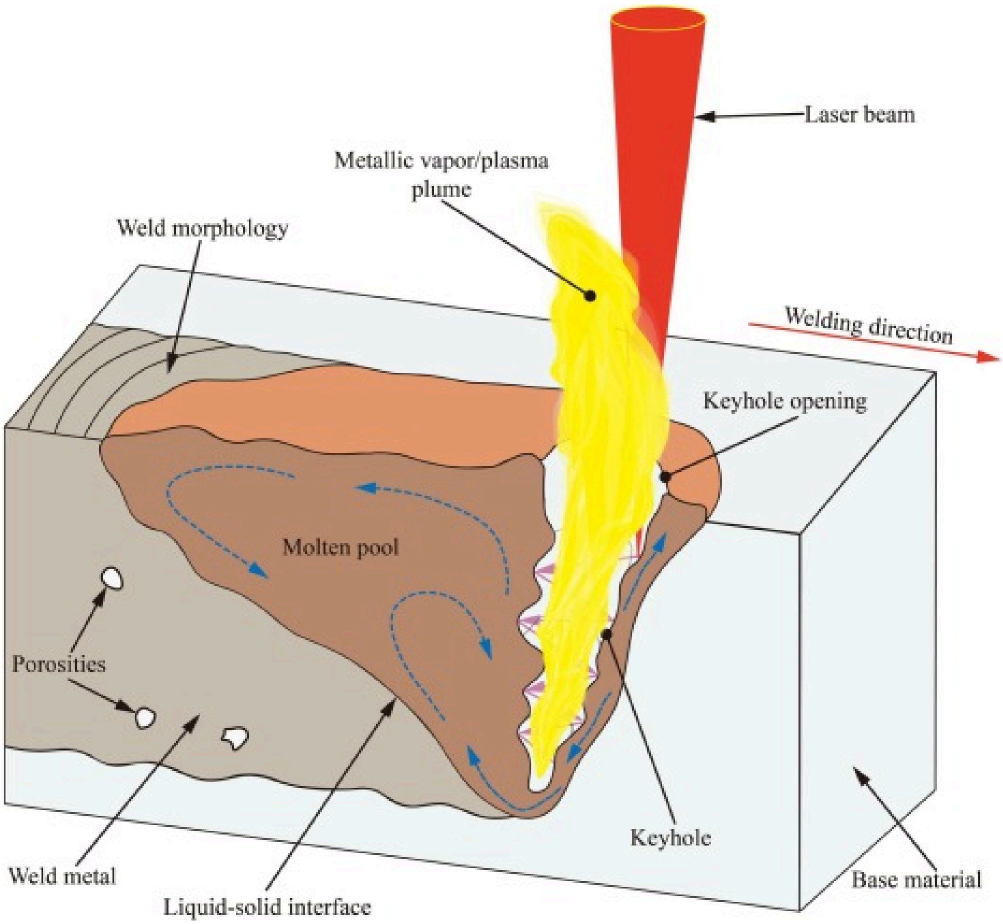

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Fig. 1. Physical phenomena in the DPLW process.

2.1. Governing equations

Numerical simulations of fluid flow behavior, as well as heat and mass transfer phenomena, are performed by solving the conservation equations of continuity, momentum, and energy [6,32].

- Continuity conservation equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \left(\rho \vec{v}\right) = S_m \tag{1}$$

where ρ denotes the density, t denotes the time, $\vec{v}$ denotes the fluid velocity, S_m represents the mass source term.

- Momentum conservation equation

$$\frac{\partial \left(\rho \vec{v}\right)}{\partial t} + \nabla \cdot \left(\rho \vec{v} \otimes \vec{v}\right) = \nabla \cdot \left(\mu \nabla \vec{v}\right) - \nabla p + \rho [1 - \beta (T - T_{ref})] \vec{g} - \frac{(1-\alpha_l)^2}{\alpha_l^3 + \zeta} A_{mush} \vec{v} + \left(\vec{F}_{st} + \vec{F}_M + \vec{P}_r\right) | \nabla \alpha_l | \frac{2\rho}{\rho_l + \rho_g} \tag{2}$$

where p denotes the pressure, μ denotes the dynamic viscosity, $\vec{g}$ denotes the gravitational acceleration, β denotes the thermal expansion coefficient, T_{ref} represents the reference temperature, α_l represents the liquid phase volume fraction with the computational cell. The term $|\nabla \alpha_l|$ is utilized to transform a surface force into a volumetric force. The subscripts l and g denote the liquid and gaseous phases, respectively. The fourth term on the right-hand side of Eq. (2) denotes the difference in fluidity caused by the liquid-solid transition. A_{mush} represents a constant representing mushy zone morphology, and ς is a small constant used to avoid division by zero. The last term represents the force exerted on the free surface, which includes the surface tension $\vec{F}_{st}$, Marangoni force $\vec{F}_M$, and recoil pressure $\vec{P}_r$.

The surface tension $\overrightarrow{F}_{st}$ is written as:

$$\overrightarrow{F}_{st} = \gamma_{st} \kappa \overrightarrow{n} \tag{3}$$

$$\gamma_{st} = \gamma_m - \frac{d\gamma}{dT}(T - T_l) \tag{4}$$

$$\overrightarrow{n} = \frac{\alpha_l}{|\nabla \alpha_l|} \tag{5}$$

where κ and $\overrightarrow{n}$ represent the interface curvature and unit vector locally normal to the interface, respectively. γ_m represents the surface tension coefficient at the melting temperature, and $\frac{d\gamma}{dT}$ represents the temperature coefficient of the surface tension.

The Marangoni force $\overrightarrow{F}_M$ is written as:

$$\overrightarrow{F}_M = \frac{d\gamma}{dT}\left[\nabla T - \left(\overrightarrow{n} \cdot \nabla T\right)\overrightarrow{n}\right] \tag{6}$$

The dominant driving force for keyhole formation is the recoil pressure $\overrightarrow{P}_r$ resulting from the intense metal evaporation. Based on previous studies [33,34], the evaporation of the molten metal surface would be confined by the ambient pressure. In the modified recoil pressure model, the recoil pressure can be regarded as the difference between the surface pressure and ambient pressure, which can be expressed as:

$$\overrightarrow{P}_r = \overrightarrow{P}_v - P_{atm} \overrightarrow{n} \tag{7}$$

$$\overrightarrow{P}_v = \begin{cases} P_{atm} \overrightarrow{n}, & 0 \le T < T_L \\ \left(a_p T^3 + b_p T^2 + c_p T + d_p\right)\overrightarrow{n}, & T_L \le T < T_H \\ \dfrac{1+\beta_R}{2} P_{atm} \exp\left[\dfrac{M_l L_v}{R T_v}\left(1 - \dfrac{T_v}{T}\right)\right]\overrightarrow{n}, & T \ge T_H \end{cases} \tag{8}$$

where $\overrightarrow{P}_v$ denotes the surface pressure, P_{atm} is the atmosphere pressure, β_R denotes the retro-diffusion coefficient, T_v is the vaporization temperature, L_v is the evaporation latent, R denotes the gas constant, M_l represents the molar mass. a_p, b_p, c_p and d_p are the fitting coefficients of a smoothed third-order polynomial. T_L and T_H represent the low and high vaporization thresholds, respectively.

• Energy conservation equation

$$\frac{\partial(\rho C_p T)}{\partial t} + \nabla \cdot \left(\rho C_p T \overrightarrow{v}\right) = \nabla \cdot (k \nabla T) + q_{laser} + (q_{rad} + q_{conv} + q_{evap}) | \nabla \alpha_l | \frac{2\rho C_p}{\rho_l C_{pl} + \rho_g C_{pg}} \tag{9}$$

where C_p and k represent the specific heat and thermal conductivity, respectively. The final term on the right-hand side of Eq. (9) denotes the rate of heat transfer occurring at the liquid-gas interface, which includes the absorbed power and the cooling rates caused by radiation, convection, and evaporation. The heat transfer caused by radiation, convection, and vaporization are given by:

$$q_{rad} = -\sigma\varepsilon\left(T^4 - T_0^4\right) \tag{10}$$

$$q_{conv} = -h_c(T - T_0) \tag{11}$$

$$q_{evap} = -L_v m_t \tag{12}$$

where T_0 represents the ambient temperature, σ, ε, and h_c denote the Stefan-Boltzmann constant, radiation emissivity, and convection coefficient, respectively. m_t denotes the mass transfer rate, which can be expressed as [32,33,35]:

$$m_t = \begin{cases} 0, & 0 \le T < T_L \\ a_m T^3 + b_m T^2 + c_m T + d_m, & T_L \le T < T_H \\ (1 - \beta_R)\sqrt{\dfrac{M_l}{2\pi RT}} P_{atm} \exp\left[\dfrac{M_l L_v}{RT_v}\left(1 - \dfrac{T_v}{T}\right)\right], & T \ge T_H \end{cases} \tag{13}$$

where a_m, b_m, c_m and d_m are the fitting coefficients of a smoothed third-order polynomial.

- Free surface tracking.

The free surface of the keyhole is tracked using the volume of fluid (VOF) method [6]:

$$\frac{\partial \alpha_l}{\partial t} + \nabla \cdot \left(\alpha_l \overrightarrow{v}\right) = \frac{m_t}{\rho} \tag{14}$$

where α_l is the fluid fraction within a computational cell ($0 \le \alpha \le 1$). The liquid-gas interface, namely where the volume fraction is between 0 and 1 ($0 < \alpha < 1$), is tracked by VOF.

2.2. Mathematical modeling of heat source

The heat source model is a crucial component in simulating the interaction between the laser ray reflection and energy absorption. The laser power density distribution is described with the following mathematical description [28]:

$$I_{LB}(x, y, z) = \frac{3P_L}{\pi R_L^2} \exp\left(-3\frac{x^2+y^2}{R_L^2}\right) \tag{15}$$

where P_L and R_L denote the laser power and focal radius of the laser spot, respectively.

In this work, the ray tracing approach is implemented to track the propagating path of laser rays inside the keyhole. Fig. 2 depicts the modeling of multiple reflections and Fresnel absorption effect within the keyhole. By incorporating the laser heat source, the laser-material interaction can be simulated within the computational framework. The laser ray reflection and energy absorption are vital factors that determine the dynamic behaviors of the keyhole (Fig. 2(a)). The continuous vertical laser beam within the calculating region (the cells within $1.6R_L$) is initially divided into groups of laser rays, carrying a specific amount of laser power (Fig. 2(b)). The initial incident direction of the laser beam is perpendicular to the metal material surface. The computational cells can be categorized into the ray-incident cell and neighboring cells according to the corresponding interface cell when calculating the absorbed energy (Fig. 2(c)). The absorbed energy of the ray-incident cell and neighboring cells are assumed to be redistributed based on the weighting function which satisfies the following equation [36].

$$w_d = \begin{cases} 1, & d \equiv (i,\ j,\ k) \\ 0.3, & d \equiv (i \pm 1,\ j,\ k),\ (i,\ j \pm 1,\ k),\ (i,\ j,\ k \pm 1) \end{cases} \tag{16}$$

where the ray-incident cell is $(i,\ j,\ k)$, the neighboring cells are $(i \pm 1,\ j,\ k)$, $(i,\ j \pm 1,\ k)$, and $(i,\ j,\ k \pm 1)$. Consequently, the absorbed energy of the computational cells can be calculated as follows:

$$e_{abs} = \begin{cases} \dfrac{e_{in}\alpha_{l,d}}{\alpha_l(i,\ j,\ k)+0.3\alpha_l(i \pm 1,\ j,\ k)+0.3\alpha_l(i,\ j \pm 1,\ k)+0.3\alpha_l(i,\ j,\ k \pm 1)}, & for\ ray-incident\ cell \\ \dfrac{0.3e_{in}\alpha_{l,d}}{\alpha_l(i,\ j,\ k)+0.3\alpha_l(i \pm 1,\ j,\ k)+0.3\alpha_l(i,\ j \pm 1,\ k)+0.3\alpha_l(i,\ j,\ k \pm 1)}, & for\ neighboring\ cells \end{cases} \tag{17}$$

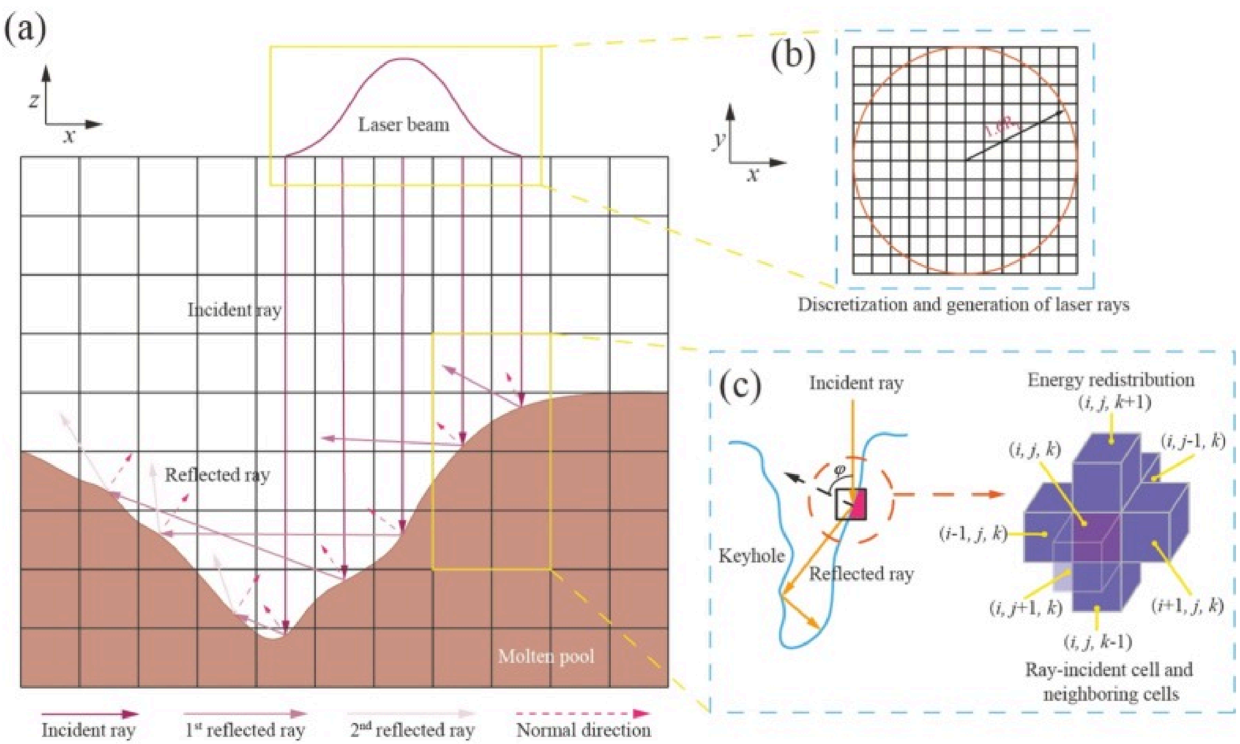

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Fig. 2. Schematic representation of heat source modeling: (a) Multiple reflections and Fresnel absorption; (b) Discretization and generation of laser rays; (c) Laser energy redistribution.

As shown in Fig. 3, the algorithm tracks the propagating path of the laser ray until it strikes the liquid-gas interface cells. Subsequently, upon reaching the reflection point, the laser ray is reflected off the liquid-gas interface. In the tracing process, the traveling trajectory of each ray is determined based on geometrical optics [36]:

$$\overrightarrow{I}_{d,r+1} = \overrightarrow{I}_{d,r} - 2\left(\overrightarrow{I}_{d,r} \cdot \overrightarrow{N}_{d,r}\right)\overrightarrow{N}_{d,r} \tag{18}$$

where $\overrightarrow{I}$ and $\overrightarrow{N}$ denote the unit vector of the laser incident direction and outward unit vector normal, respectively. The subscripts d and r represent the ray identity number and reflection index, respectively. To ensure numerical accuracy, the ray tracing algorithm terminates when laser rays exit the computational domain or when the remaining energy decreases to 1% of the original energy. In terms of each laser ray incidence, the energy of the incident laser ray is partially absorbed by the keyhole wall. The absorptivity is determined by [37,38]:

$$R_a = 1 - \frac{R_s + R_p}{2} \tag{19}$$

where R_s and R_p denote the reflectivity corresponding to the parallel-polarized and perpendicular-polarized light, respectively.

$$R_s = \frac{\chi^2+\eta^2-2\chi\cos\varphi+\cos^2\varphi}{\chi^2+\eta^2+2\chi\cos\varphi+\cos^2\varphi} \tag{20}$$

$$R_p = R_s\frac{\chi^2+\eta^2-2\chi\cos\varphi\tan\varphi+\sin^2\varphi\tan^2\varphi}{\chi^2+\eta^2+2\chi\cos\varphi\tan\varphi+\sin^2\varphi\tan^2\varphi} \tag{21}$$

where φ denotes the incident angle, χ and η are the functions of the refractive index and extinction coefficient, respectively. Considering the refraction index $\widetilde{n} = n - i\tau$, χ and η can be calculated as follows:

$$2\chi^2 = \sqrt{\left(n^2-\tau^2-\sin^2\varphi\right)^2+4n^2\tau^2}+\left(n^2-\tau^2-\sin^2\varphi\right) \tag{22}$$

$$2\eta^2 = \sqrt{\left(n^2-\tau^2-\sin^2\varphi\right)^2+4n^2\tau^2}-\left(n^2-\tau^2-\sin^2\varphi\right) \tag{23}$$

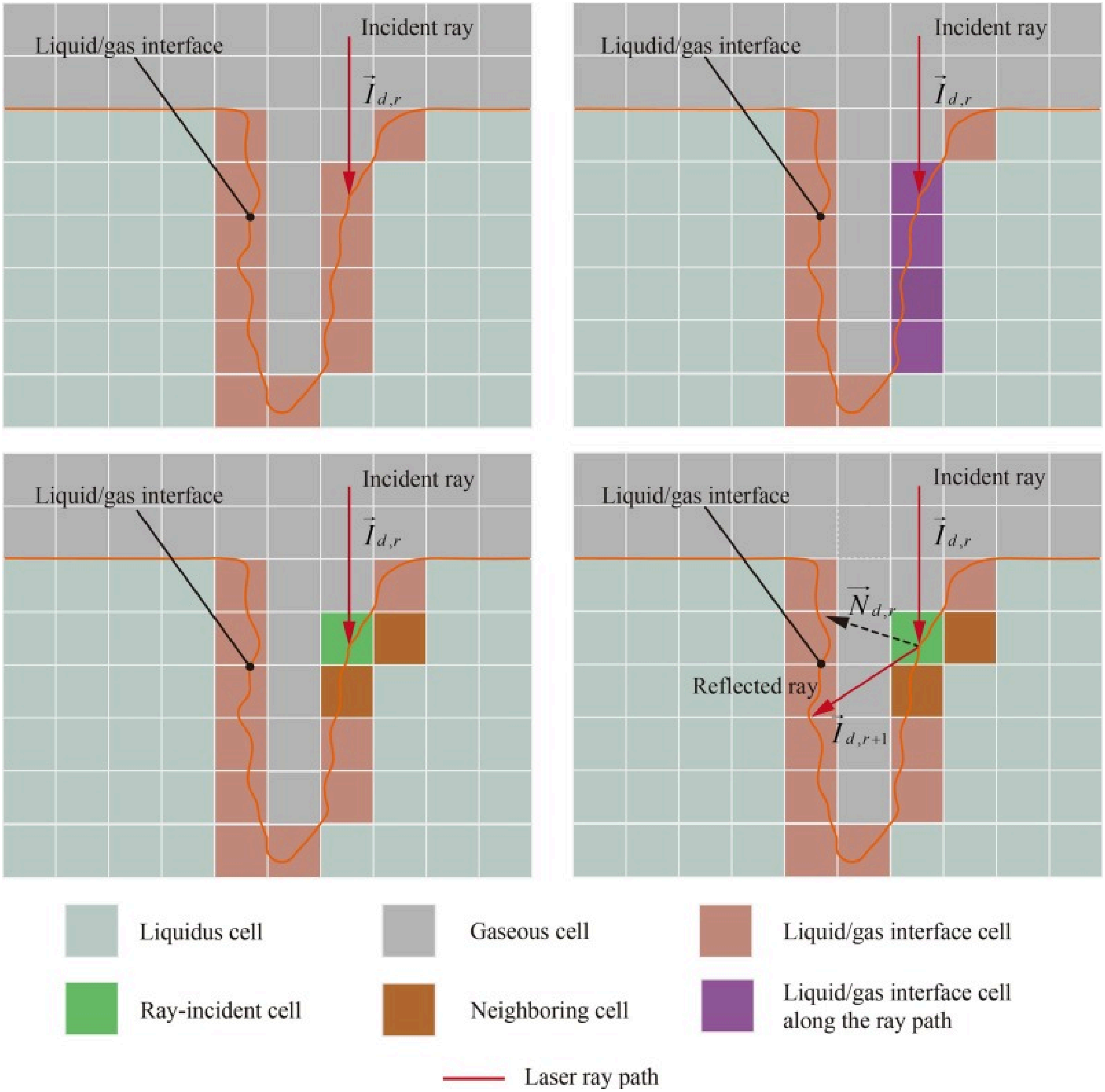

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Fig. 3. Schematic representation of tracing process of laser ray and cell distribution.

n and τ can be expressed as by the relative electric permittivity $\tilde{\varepsilon} = \varepsilon_r - i\varepsilon_i$:

$$n = \frac{1}{\sqrt{2}}\left[\left(\varepsilon_r^2 + \varepsilon_i^2\right)^{1/2} + \varepsilon_r\right]^{1/2} \tag{24}$$

$$\tau = \frac{1}{\sqrt{2}}\left[\left(\varepsilon_r^2 + \varepsilon_i^2\right)^{1/2} - \varepsilon_r\right]^{1/2} \tag{25}$$

$$\varepsilon_r = 1 - \frac{\omega_p^2}{\zeta^2 + \delta^2} \tag{26}$$

$$\varepsilon_i = \frac{\delta}{\zeta}\frac{\omega_p^2}{\zeta^2 + \delta^2} \tag{27}$$

where ω_p denotes the plasma frequency, ζ denotes the laser frequency, δ denotes the damping frequency.

$$\omega_p = \sqrt{\frac{N_e e^2}{m_e \varepsilon_0}} \tag{28}$$

$$f_L = 2\pi\frac{c}{\lambda} \tag{29}$$

$$\delta = \frac{N_e e^2}{m_e}\rho_r(T) = \omega_p^2 \varepsilon_0 \rho_r(T) \tag{30}$$

where N_e, e, m_e, and ε_0 denote the average electron density, the absolute value of the elementary charge, the electron mass, the vacuum permittivity, respectively. c and λ are the light speed and laser wavelength, respectively. ρ_r represents the temperature-dependent electrical resistivity, leading to a damping frequency that depends strongly on the fluid temperature [38]. The average electron density N_e is determined as the atomic-fraction-weighted mean of the constituent elements.

The metallic vapor plume generated contains plenty of the metallic particles generated in laser welding process. The resulting laser attenuation through scattering and absorption can be expressed as follows [39]:

$$I(d_{dis})= I_{in}[1-\exp(-(\beta_{sca}+\beta_{abs})d_{dis})] \tag{31}$$

$$\beta_{sca} = \frac{8}{3}\left(\frac{2\pi r_m}{3}\right)^4 \mid \frac{m_p^2-1}{m_p^2+2} \mid \times N\pi r_m^2 \tag{32}$$

$$\beta_{sca} = -\frac{8\pi r_m}{\lambda}\mathrm{Im}\left(\frac{m_p^2-1}{m_p^2+2}\right)\times N\pi r_m^2 \tag{33}$$

where N and r_m are the particle density and radius, respectively. d_{dis} is the traveling distance of the laser ray, m_p denotes the complex refractive index of the particle.

2.3. Simulation setup and material properties

Fig. 4 presents the schematic diagram of the computational domain for the numerical simulation. The three-dimensional domain had dimensions of 4 mm×3 mm×4 mm, which was discretized by the uniform grid size of 0.05 mm. The combination of the totalPressure condition on pressure and pressureInletOutletVelocity on velocity was defined at the gaseous faces (EFJI, FGKJ, GHLK, HEIL, and IJKL). No-slip and zero-gradient boundary conditions for velocity and pressure were applied at the metallic faces (ABFE, BCGF, CDHG, DAEH, and ABCD). Convective boundary conditions were imposed on all surfaces of the computational domain. The initial temperature and reference pressure were set to 300 K and 101,325 Pa, respectively.

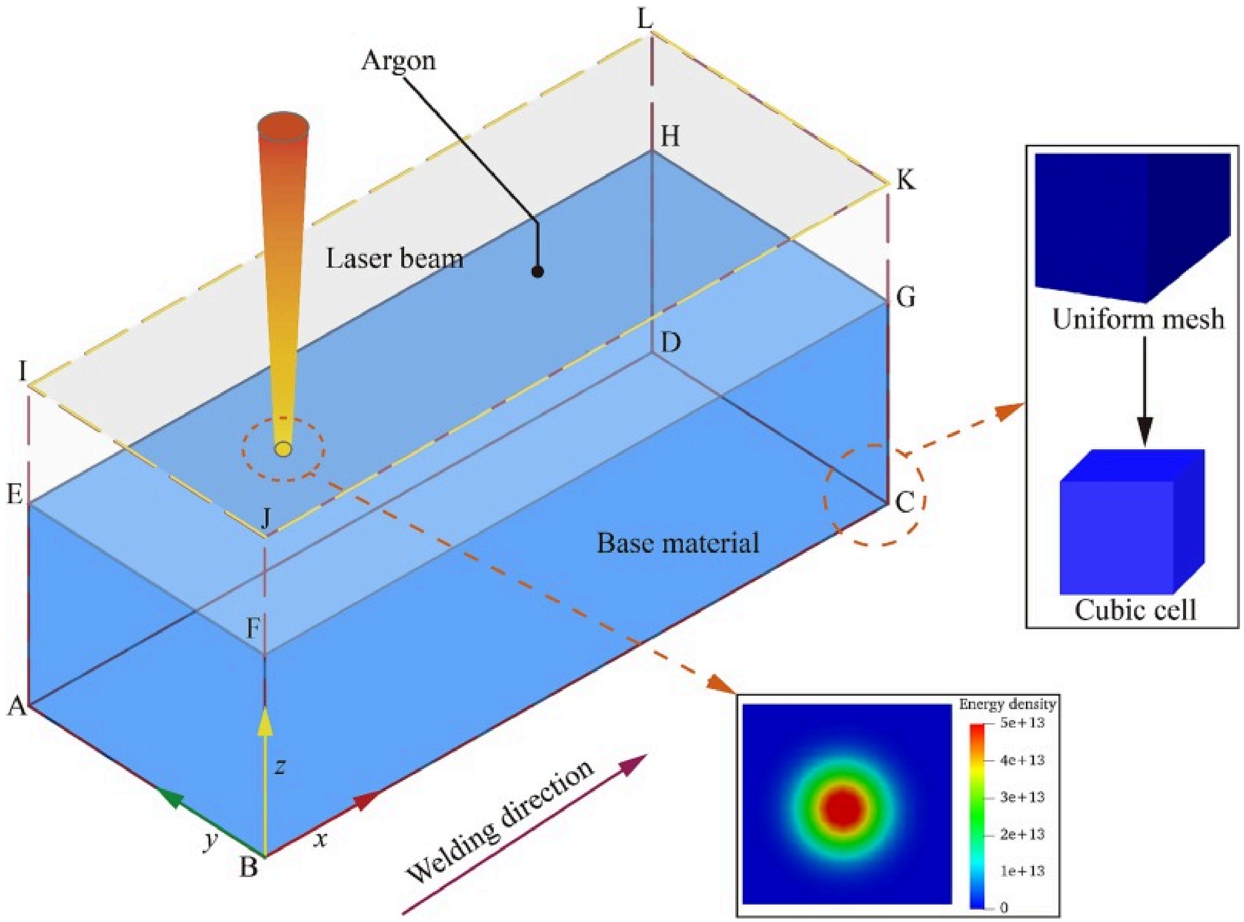

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Fig. 4. Schematic representation of the computational domain.

The numerical model was implemented to solve the high-resolution thermal field using the open-source library OpenFOAM. The relevant governing equations were modified in lasermeltingFoam solver, which was a modified version of laserbeamFoam solver [40]. The transient pressure-velocity coupling was resolved through Pressure-Implicit with Splitting of Operators (PISO) by sequentially solving Eqs. (14), (1), (2), (9) at every temporal iteration. After each computational time step, the multiple reflections of the laser rays at the free surface were recorded to calculate the laser absorptivity. In this work, numerical results of the keyhole behavior and laser absorption were saved at time intervals of 5×10^{-6} s and 5×10^{-7} s, respectively. To ensure numerical stability, the Courant–Friedrichs–Lewy (CFL) number was maintained at 0.4. Table 1, Table 2 list the thermophysical properties of Ti6Al4V and argon gas, computational constants, vaporization constants, and related parameters employed in the physics-based absorption model. The simulations were performed on a high-performance computer with 104 Intel-Xeon and 192 GB RAM.

Table 1. Thermophysical parameters of Ti6Al4V used for simulations [6,32,36].

Nomenclature	Symbol	Unit	Value
Liquid density	ρ_l	$kg{\cdot}m^{-3}$	4400
Thermal conductivity of solid phase	k_s	$W{\cdot}m^{-1}{\cdot}K^{-1}$	13
Thermal conductivity of liquid phase	k_l	$W{\cdot}m^{-1}{\cdot}K^{-1}$	33
Special heat of solid phase	C_{ps}	$J{\cdot}kg^{-1}{\cdot}K^{-1}$	573
Special heat of liquid phase	C_{pl}	$J{\cdot}kg^{-1}{\cdot}K^{-1}$	750
Dynamic viscosity	μ	$kg{\cdot}m^{-1}{\cdot}s^{-1}$	5×10^{-3}
Solidus temperature	T_s	K	1878
Liquidus temperature	T_l	K	1928
Vaporization temperature	T_v	K	3533
Latent heat of fusion	L_f	$J{\cdot}kg^{-1}$	2.86×10^5
Latent heat of vaporization	L_v	$J{\cdot}kg^{-1}$	9.7×10^6
Surface tension coefficient	γ_m	$N{\cdot}m^{-1}{\cdot}K^{-1}$	1.63
Temperature coefficient of surface tension	$d\gamma/dT$	$N{\cdot}m^{-1}$	-2.8×10^{-4}

Nomenclature	Symbol	Unit	Value
Thermal expansion coefficient	β	K^{-1}	2.5×10^{-5}
Convection coefficient	h_c	$W\cdot m^{-2}\cdot K^{-4}$	20
Emissivity	ε	–	0.3
Stefan-Boltzmann constant	σ	$W\cdot m^{-2}\cdot K^{-4}$	5.67×10^{-8}

Table 2. Thermophysical properties of argon gas, computational constants, vaporization constants, and related parameters used in physics-based absorption model [37,41].

Nomenclature	Symbol	Unit	Value
Molar mass (Ar)	M_g	$kg\cdot mol^{-1}$	0.03995
Specific heat (Ar)	C_{pg}	$J\cdot kg^{-1}\cdot K^{-1}$	520
Thermal conductivity (Ar)	k_g	$W\cdot m^{-1}\cdot K^{-1}$	1.7×10^{-2}
Dynamic viscosity (Ar)	μ_g	$kg\cdot m^{-1}\cdot s^{-1}$	2.1×10^{-5}
Mushy zone constants	A_{mush}	$kg\cdot m^{-3}\cdot s^{-1}$	2×10^{8}
	ζ	–	5.67×10^{-8}
Atmospheric pressure	P_{atm}	Pa	101,325
Gas constant	R	$J\cdot K^{-1}\cdot mol^{-1}$	8.314
Ambient temperature	T_0	K	300
Retro-diffusion coefficient	β_R	–	0.18
Low vaporization threshold	T_L	K	3300
High vaporization threshold	T_H	K	3900

Nomenclature	Symbol	Unit	Value
Mass transfer rate constants	a_m/b_m	–	$1.27038\times10^{-7}/-8.41341\times10^{-4}$
	c_m/d_m	–	$1.40570/-4.19826\times10^{1}$
Surface pressure constants	a_p/b_p	–	$1.12712\times10^{-3}/-1.14524\times10^{1}$
	c_p/d_p	–	$3.88039\times10^{4}/-4.38415\times10^{7}$
Electron mass	m_e	kg	9.11×10^{-31}
Elementary charge	e	C	1.60×10^{-19}
Speed of light	c	$m\cdot s^{-1}$	3×10^{8}
Vacuum permittivity	ε_0	$F\cdot m^{-1}$	8.85×10^{-12}
Free electron density	N_e	m^{-3}	2.20×10^{29}
Laser wavelength	λ	nm	1064

2.4. Statistical analysis of simulation results

In this work, kernel density estimation (KDE) is employed to evaluate the stochastic characteristics of the keyhole dynamics and laser energy absorption in the DPLW process. As a non-parametric statistical approach, KDE is used to analyze the probability density function (PDF) of the continuous random variable. Unlike parametric methods that assume a specific distribution for the data, KDE does not require any prior assumption about the form of the underlying distribution. Assuming a dataset $X=\{x_1,x_2,\cdots x_n\}$, the KDE at the location x is given by [42]:

$$\widehat{f}_n(x)=\frac{1}{n_s h}\sum_{i=1}^{n_s}K\left(\frac{x-x_i}{h}\right) \tag{34}$$

where $\widehat{f}_n$ denotes the estimated PDF, n_s denotes the sample number, h represents the bandwidth, K represents the kernel function. The Gaussian kernel is chosen to perform kernel density estimation:

$$K(u)= \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{u^2}{2}\right) \tag{35}$$

Laser deep penetration welding is a typical nonlinear dynamic system, which is characterized by complex thermal and fluid dynamic interactions. While the nonlinear dynamical system exhibits unstable and stochastic characteristics over short time scales, it eventually converges toward an equilibrium or relatively stable state over extended period [[43], [44], [45], [46]]. As the system deviates from the equilibrium point, the keyhole experiences the dramatic variations. These variations originate from the intricate coupling among laser energy, keyhole dynamics, and molten pool dynamics [15,47]. The fluctuations, which often exhibit periodic characteristics, can degrade welding stability and result in quality issues. Therefore, the nonlinear nature of the laser welding process can be better elucidated through an in-depth assessment of keyhole dynamics and laser absorption. In the KDE curve, the peak probability density identifies the area where data points are more densely concentrated, reflecting a higher probability density. Accordingly, higher KDE values suggest increased probability of an event occurring within the corresponding region. In this work, the characteristic value at the peak probability density is defined as the equilibrium state point.

3. Experimental setup

Fig. 5 depicts the schematic representation of the experimental setup for the DPLW process. The bead-on-plate (BOP) experiments were carried out by utilizing the Nd:YAG laser (wavelength: 1064 nm). Welding experiments were performed on Ti6Al4V titanium alloy plate (thickness: 3 mm). The process parameters for the DPLW experiments are summarized in Table 3. A parabolic mirror with a focal length of 160 mm was employed to focus the Nd:YAG laser beam on the top surface of the base material. The radius of the focused laser beam is 0.3 mm. All experiments were conducted under protective atmosphere conditions. Argon shielding gas was supplied at a flow rate of 15 L/min to protect the top surface of the weld from atmospheric contamination. Additionally, a backside protection device was implemented to isolate the molten material at the lower surface from potential contamination in laser welding process. Argon was continuously delivered to the backside protection device at a flow rate 25L/min through the dedicated inlet.

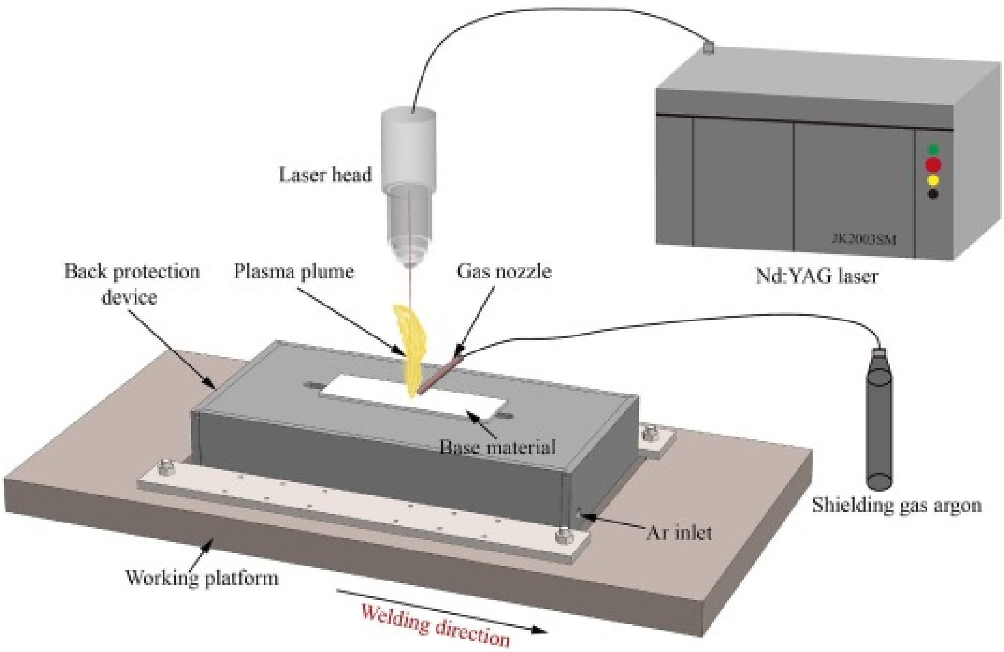

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Fig. 5. Schematic diagram of the experimental setup for the DPLW experiments.

Table 3. Process parameters for welding experiments.

Case	Laser power (W)	Welding speed ($mm{\cdot}s^{-1}$)
A	1200	30
B	1300	30

Case	Laser power (W)	Welding speed ($mm \cdot s^{-1}$)
C	1400	30

Prior to welding experiments, the Ti6Al4V plate surface was mechanically polished using sandpaper and subsequently cleaned with anhydrous ethanol. After welding, the weld specimens were sequentially cut, ground, polished, and finally etched in a solution of H_2O:HF:HNO_3=16:1:3. Metallographic observation was performed using optical microscopy to reveal the fusion zone morphology through cross-sectional weld analysis. Simultaneously, longitudinal cross-sections of the welds were obtained to facilitate analysis of weld formation characteristics in the DPLW process.

4. Results and discussion

4.1. Numerical model verification

To verify the established model, the experimental cases (Table 3) were simulated, and the computed results were evaluated against the experimental data. Fig. 6 illustrates the differences in weld shape and cross-sectional dimensions obtained from both experiments and simulations under varying welding conditions. The predicted weld penetration depth aligns well with the experimental measurements. It is noted that there is a discrepancy between the simulation and experimental results in the width of the fusion line, which is mainly attributed to the omission of plasma thermal radiation effects in the model [48]. Therefore, the validity of the model is supported by the agreement between numerical simulations and experimental results. The observed consistency confirms that the developed model offers a reliable foundation for further investigation into keyhole dynamics, laser absorption within the keyhole, and the transient coupling between them in the DPLW process.

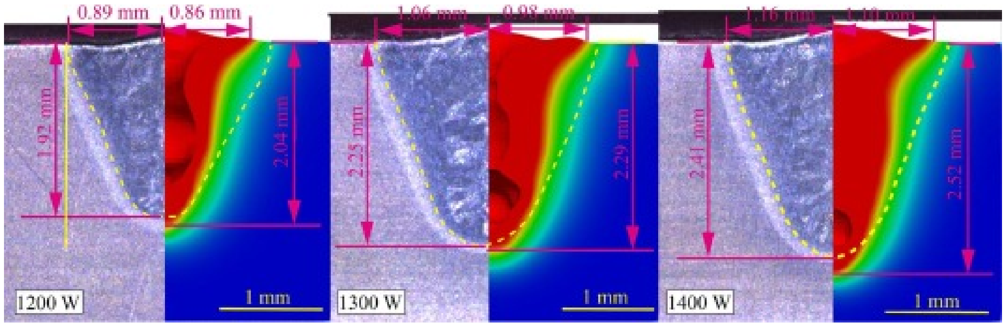

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Fig. 6. Analysis of weld profiles from experiments and simulations.

4.2. Evaluation and analysis of keyhole dynamics

Numerical simulation can offer quantitative information on the transient behavior of the keyhole to facilitate further understanding and analysis of the DPLW process. Laser welding is a highly stochastic, time-varying, and nonlinear process influenced by multiple interacting factors. Further investigations are required into the variation and distribution characteristics of the morphology parameters including the keyhole depth and keyhole width, as illustrated in Fig. 7.

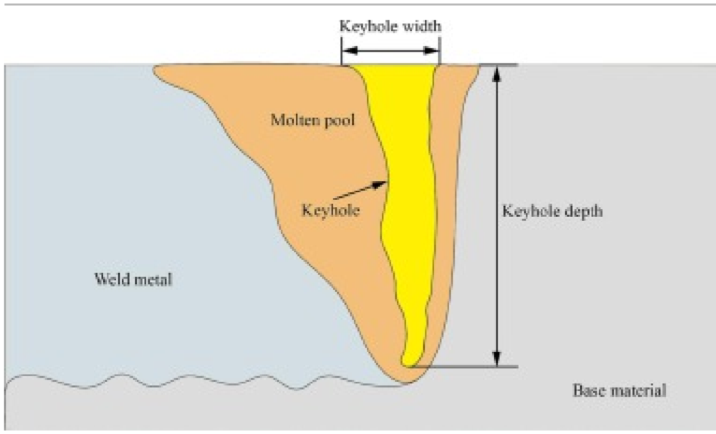

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Fig. 7. Characteristic parameters for evaluating dynamic keyhole behavior.

To reveal the keyhole dynamics in the DPLW process, the evolution process of the predicted keyhole depth is depicted in Fig. 8(a). As revealed by the numerical results, the variation in keyhole depth can be categorized into three distinct phases: (I) rapid increase stage; (II) gradual increase and initial fluctuation stage; (III) keyhole depth fluctuation stage. Fig. 8(b) presents a magnified view of the keyhole depth evolution between 25 ms and 45 ms. Numerical results reveal that, at all laser power levels, the keyhole exhibits periodic fluctuations during its quasi-steady state due to continuous interaction between the laser beam and the metal material. Furthermore, with higher laser powers resulting in more pronounced and rapid fluctuations, the keyhole depth is significantly influenced by the laser power. To further investigate the characteristics of weld formation in the DPLW process, the longitudinal cross-sections of the welds is examined, as shown in Fig. 9. Compared to the dramatic fluctuations in the keyhole depth, the fluctuations in the weld penetration depth exhibits relatively small amplitude variations. It is noteworthy that the number of the residual pores shows a significant increasing trend with the increasing laser power, which corresponds to the phenomenon that the increasing laser power leads to more intense keyhole fluctuations. Additionally, the pores are predominantly distributed in the middle-lower region of the weld longitudinal cross-section. This spatial distribution pattern indicates that pore formation is closely related to the dynamic behavior of the keyhole. The experimental observations in Fig. 9 provide important indirect evidence for the periodic fluctuation behavior of the keyhole. Therefore, the experimental results in Fig. 9 further validate that the numerical model established in this work can effectively capture the behavior characteristics of the molten pool and keyhole in the DPLW process.

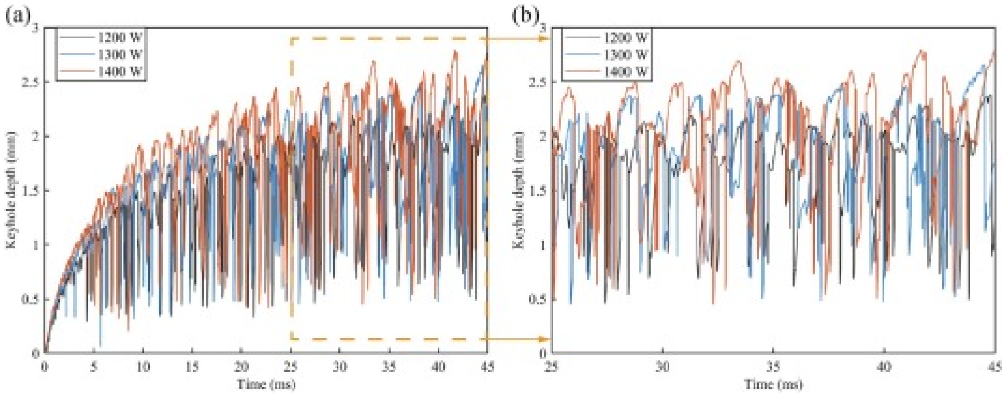

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Fig. 8. Evolution process of the keyhole depth at different laser power levels: (a) Temporal evolution of the predicted keyhole depth throughout the entire simulation process; (b) Predicted variations in the keyhole depth from 25 ms to 45 ms.

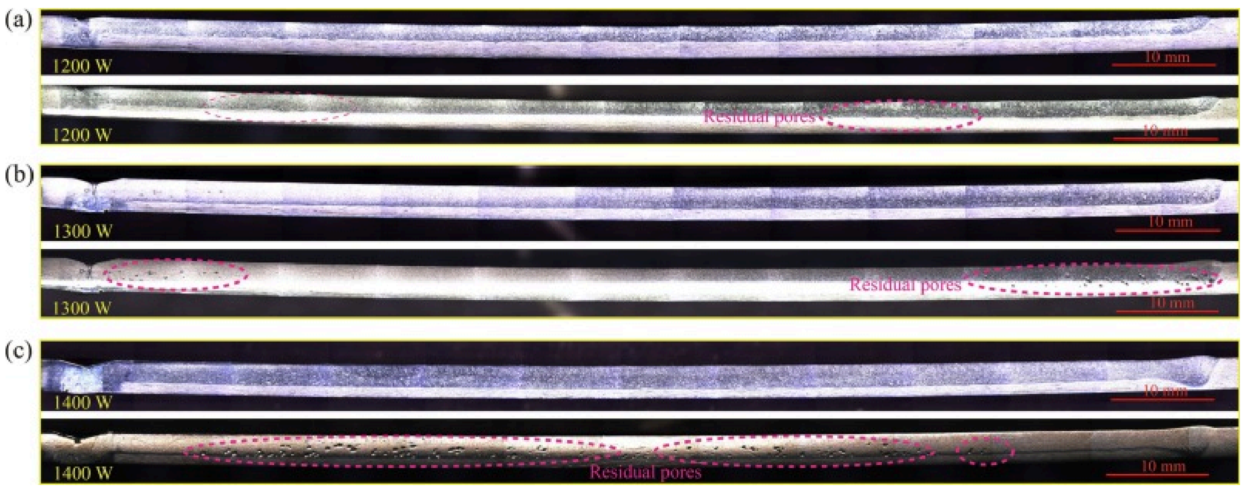

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Fig. 9. Longitudinal cross-sectional morphology of the welds and porosity distribution at different laser power levels: (a) 1200W; (b) 1300W; (c) 1400W.

KDE is employed to evaluate the probability density distribution characteristics of the keyhole depth in the fluctuation stage (25–45ms). Fig. 10 illustrates the histogram and probability density curve for the predicted keyhole depth. The histogram represents the discrete distribution of the keyhole depth, while the probability density curve provides a continuous estimation of the underlying distribution. The keyhole depth distribution, as represented by the KDE curve, shows a unimodal pattern. Furthermore, the keyhole depth distribution exhibits a left-skewed normal pattern, suggesting that most data points are concentrated on higher depth values. As the laser power increases, a rightward shift in the peak position of the kernel density curve is observed, suggesting an increase in the equilibrium depth. Higher laser power generates deeper keyholes with correspondingly wider fluctuation ranges. The position of the peak corresponds to the equilibrium state point of the keyhole during the fluctuation stage, representing the most frequent and stable keyhole depth. In the DPLW process, the equilibrium state refers to the condition where the keyhole depth remains relatively stable over time [49]. In this case, the minor deviations occurring around this equilibrium state point can be interpreted as a stable process. Conversely, the non-equilibrium state occurs when the keyhole significantly departs from its stable configuration. In the non-equilibrium state, the stability issue of the keyhole becomes prominent. The keyhole collapse is a typical case of structural instability when the keyhole significantly deviates from its equilibrium state. As shown Fig. 10, the left portion of the distribution corresponds to instances of rapid depth reduction associated with keyhole collapse. Therefore, the statistical analysis of keyhole dynamic states using KDE can offers valuable insights into the stability issues of the DPLW process.

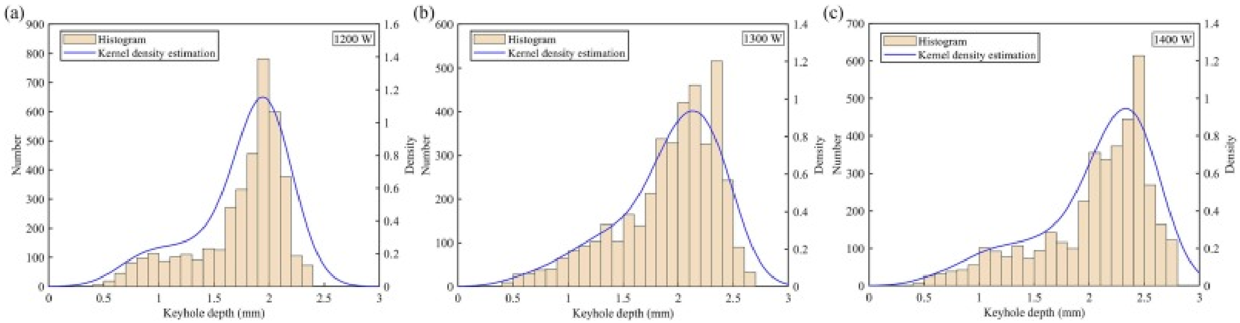

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Fig. 10. Histogram and KDE curve for the keyhole depth at different laser power levels: (a) 1200W; (b) 1300W; (c) 1400W.

Fig. 11 illustrates the time-dependent behavior of the keyhole opening width. The results reveal the keyhole width undergoes cyclic fluctuations characterized by alternating stages of contraction and expansion. Notably, greater fluctuation amplitudes are observed at higher laser power levels. This trend indicates that higher laser power levels result in increased keyhole instability and greater fluctuations in width. Fig. 12 shows that the histogram and corresponding probability density distribution for the keyhole width. The KDE curve of the keyhole width fluctuations approximately follows a normal distribution pattern. The equilibrium widths at laser power levels of 1200W, 1300W, and 1400W are 0.72mm, 0.73mm, and 0.76mm, respectively. Accordingly, higher laser power contributes to a noticeable rightward shift in the mode of the kernel density distribution for keyhole width.

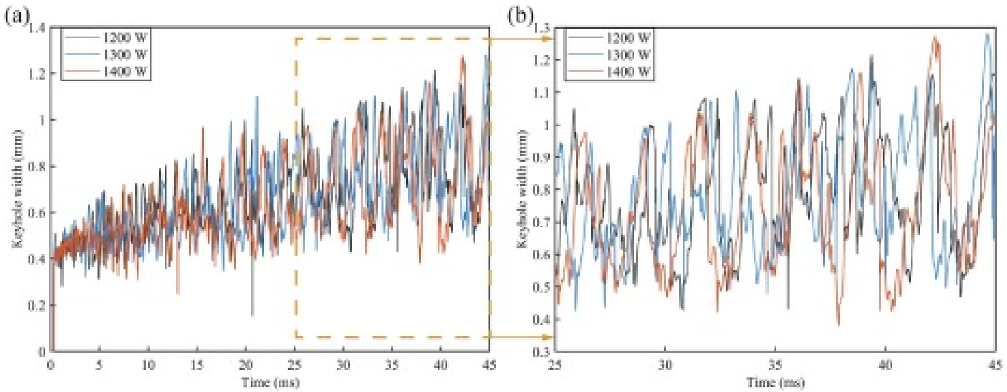

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Fig. 11. Temporal evolution of the keyhole width at different laser power levels: (a) Temporal evolution of the predicted keyhole width throughout the entire simulation process; (b) Predicted variations in the keyhole width from 25 ms to 45 ms.

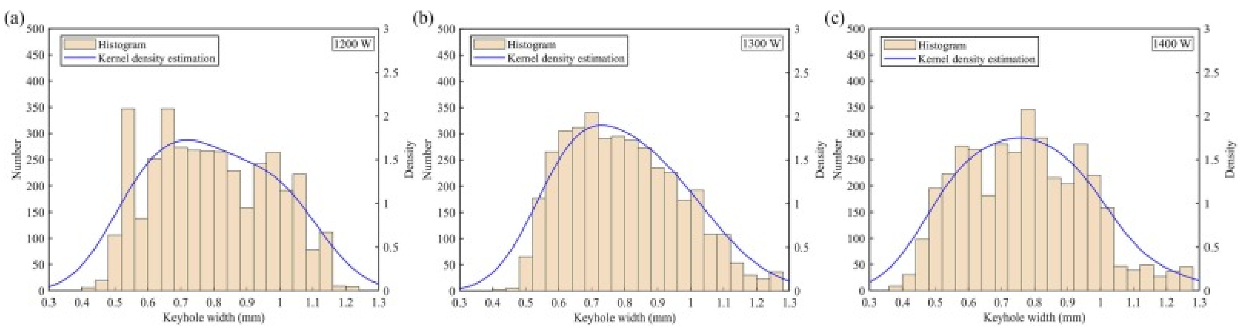

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Fig. 12. Histogram and KDE curve for the keyhole width at different laser power levels: (a) 1200W; (b) 1300W; (c) 1400W.

The static model of keyhole dynamics can serve as a theoretical foundation for understanding the trend in width variation. [44,45]. As shown in Fig. 13(a), to ensure the stable keyhole configuration, the pressure on the keyhole wall must be balanced according to the following equations [50]:

$$\Delta p = p_{abl} + \Delta p_g - p_\gamma \tag{36}$$

$$p_{abl} = m n_g u_g^2 \tag{37}$$

$$\Delta p_g = \frac{1}{3} Cg(Re) \left(\frac{d_k}{r_k} \right)^2 p_{abl} \tag{38}$$

$$p_\gamma = \frac{\gamma}{r_k} \tag{39}$$

where p_{abl}, Δp_g, and p_γ represent the ablation pressure, excessive pressure, and surface tension, respectively. r_k and d_k denote the keyhole radius and depth, respectively. m represents the atomic mass of the metallic species, n_g and u_g denote the hydrodynamic density and motion characteristics of the metallic vapor plume within the keyhole, respectively. $Cg(Re)$ depends on the Reynolds number, and γ represents the surface tension coefficient. The keyhole undergoes contraction and expansion when the pressure deviation Δp fluctuates (Fig. 13(b)). Fig. 13(c) illustrates the pressure stability intersection curve for the keyhole. The variation curve of the resultant pressure for p_{abl} and Δp_g under higher laser power is positioned above that under lower laser power. The dynamic state of the keyhole features a stable equilibrium point (B and B') and an unstable equilibrium point (A and A'). The resulting pressure deviation drives the keyhole to fluctuate around the stable equilibrium point (Fig. 13(d)). In contrast, the unstable equilibrium point lacks the ability to resist the fluctuation of the keyhole wall. An increase in laser power results in a more pronounced pressure imbalance, which expands the separation between the steady and unsteady equilibrium points. Therefore, higher laser power leads to a larger fluctuation range for the stable fluctuation, which is consist with the result shown in Fig. 12. The above analysis indicates that statistical evaluation of the keyhole behavior effectively supports the applicability of the quasi-static model describing keyhole dynamics.

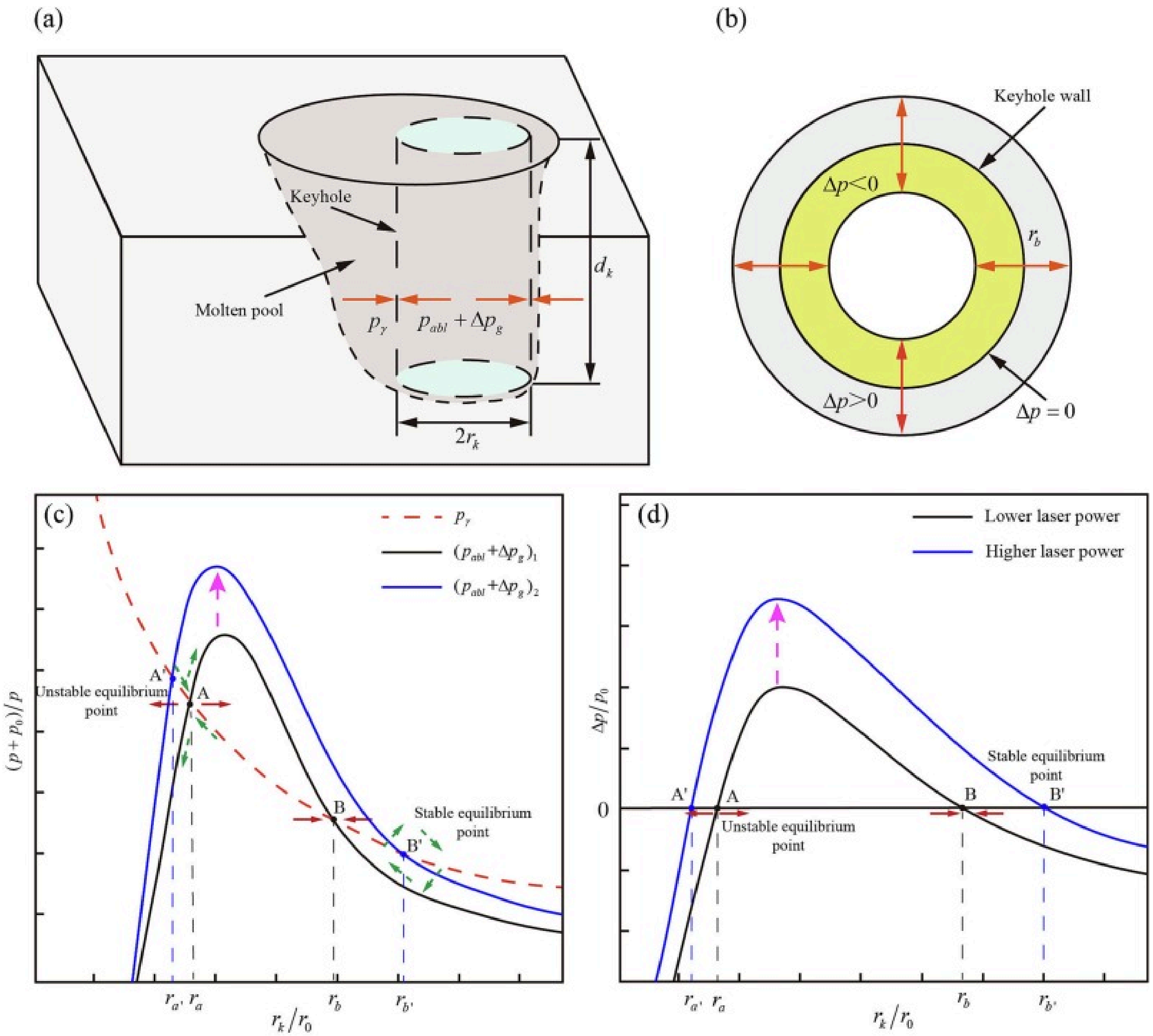

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Fig. 13. Pressure balance for stable keyhole morphology: (a) Static model of pressure balance condition; (b) Keyhole radius fluctuation driven by the pressure deviation; (c) Analysis of pressure balance at different laser power levels; (d) Analysis of the pressure deviation Δp at different laser power levels.

4.3. Evaluation and analysis of laser energy absorption for dynamic keyhole

The coupling mechanism between laser energy absorption and keyhole dynamics has not been quantitatively analyzed in the previous studies. This limitation arises from the challenges associated with accurately quantifying laser energy absorption using current experimental techniques. In this section, a quantitative investigation of laser absorption is conducted by simulating the interaction between the keyhole wall and laser beam.

As illustrated in Fig. 14, the time-resolved laser absorptivity for the entire keyhole can be obtained by the ray tracing approach. The slender keyhole can be observed under all simulation parameters employed, resulting in similar characteristic patterns in laser energy absorption evolution. Consistent with the keyhole depth evolution, the laser energy absorption process can also be categorized three characteristic stages. During the fluctuation stage, the laser absorptivity within the keyhole closely aligns with previous research findings [51]. The keyhole formation is responsible for the sharp enhancement in laser energy absorption observed during the initial process. The formation of the keyhole, driven by rapid metal vaporization, facilitates enhanced laser energy absorption through multiple internal reflections. The absorption behavior of the laser beam is highly keyhole morphology dependent. Therefore, as the keyhole morphology evolves dynamically over time, the periodic fluctuations of the absorption can be observed at all laser powers levels in the steady fluctuation stage.

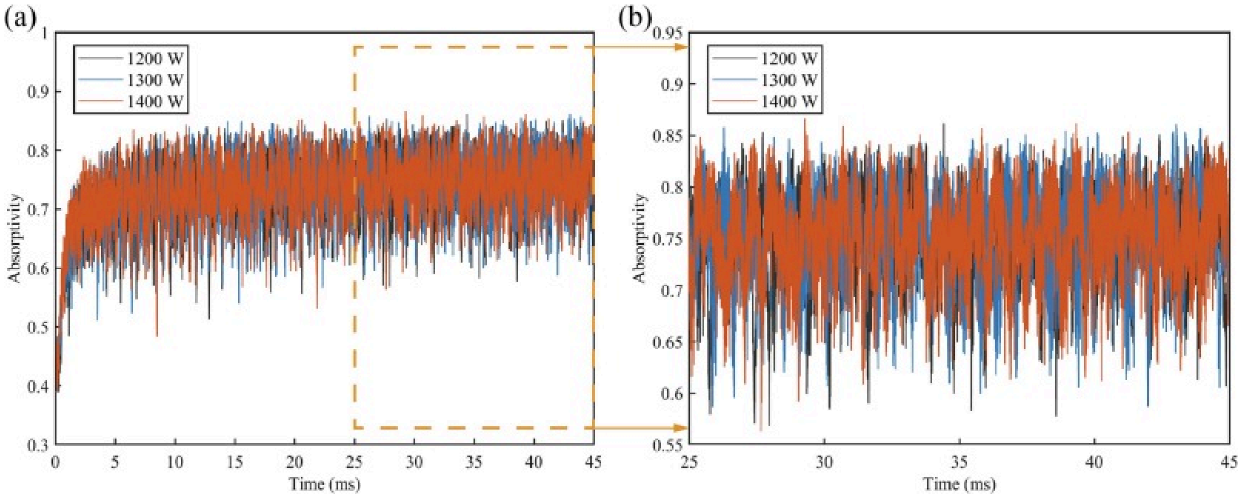

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Fig. 14. Evolution process of instantaneous laser absorptivity at different laser power levels: (a) Time-resolved laser absorptivity throughout the entire simulation process; (b) Predicted variations in the laser absorptivity from 25 ms to 45 ms.

Notably, the laser beam interacts with the metallic vapor/plasma plume prior to entering the keyhole. Consequently, the energy available for absorption within the keyhole corresponds to the rest energy of the laser rays after traveling through the metallic vapor/plasma plume in this work. KDE is employed to analyze the laser absorptivity for the entire keyhole in the fluctuation stage. Fig. 15 shows the histogram and corresponding KDE curve of the laser absorptivity for the entire keyhole. The laser absorptivity distribution for the entire keyhole can be well described by a normal probability density function. The overall trend indicates that the variation range of the laser absorptivity broadens with increasing laser power, while the peak of the distribution remains consistently centered around 0.75.

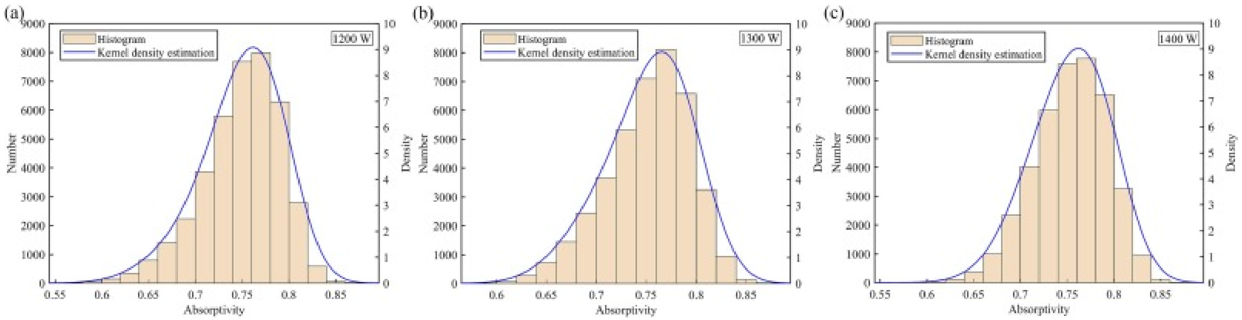

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Fig. 15. Histogram and KDE curve of the absorptivity for the entire keyhole at different laser power levels.

For a more detailed examination of the laser absorptivity in the fluctuation stage, the total absorbed laser absorption is decomposed into contributions from the front and rear keyhole walls. Fig. 16 shows the quantitative dynamic data of the laser energy absorption on the front and rear walls. As presented in Fig. 16(a), the vertical plane including the lowest point of the keyhole separates the keyhole into the front and rear parts. Numerical results highlight the spatial non-uniformity and temporal variability of energy deposition within the keyhole (Fig. 16(b) and (d)). Fig. 16(c) and (e) show that laser energy absorbed by the front and rear surfaces of the keyhole exhibits a clear periodic pattern. Fig. 17 presents a statistical comparison of laser energy absorbed by the front and rear surfaces of the keyhole at different laser power levels. The probability density functions of the laser absorptivity on both the front and rear keyhole surfaces demonstrate approximately normal distribution patterns. Table 4 summarizes the laser energy absorption characteristics corresponding to each simulation case. The equilibrium value of the laser absorptivity on the front and rear keyhole surface are 40% and 35%, respectively, reflecting an asymmetry in laser energy deposition. Additionally, Fig. 14, Fig. 16 indicates that compared to the front wall, the laser absorptivity across the entire keyhole and on its rear wall exhibits noticeable differences. Specifically, the rear part of the keyhole and the overall keyhole profile exhibit more pronounced fluctuations in the laser absorptivity than the front surface. Considering that the laser-material interaction is highly dependent on keyhole morphology, the observed fluctuations in the laser absorptivity indicate that the rear keyhole wall undergoes dramatic shape variations. These findings emphasize the significance of controlling the keyhole morphology and stability to enhance laser energy utilization and improve the weld quality in the DPLW process.

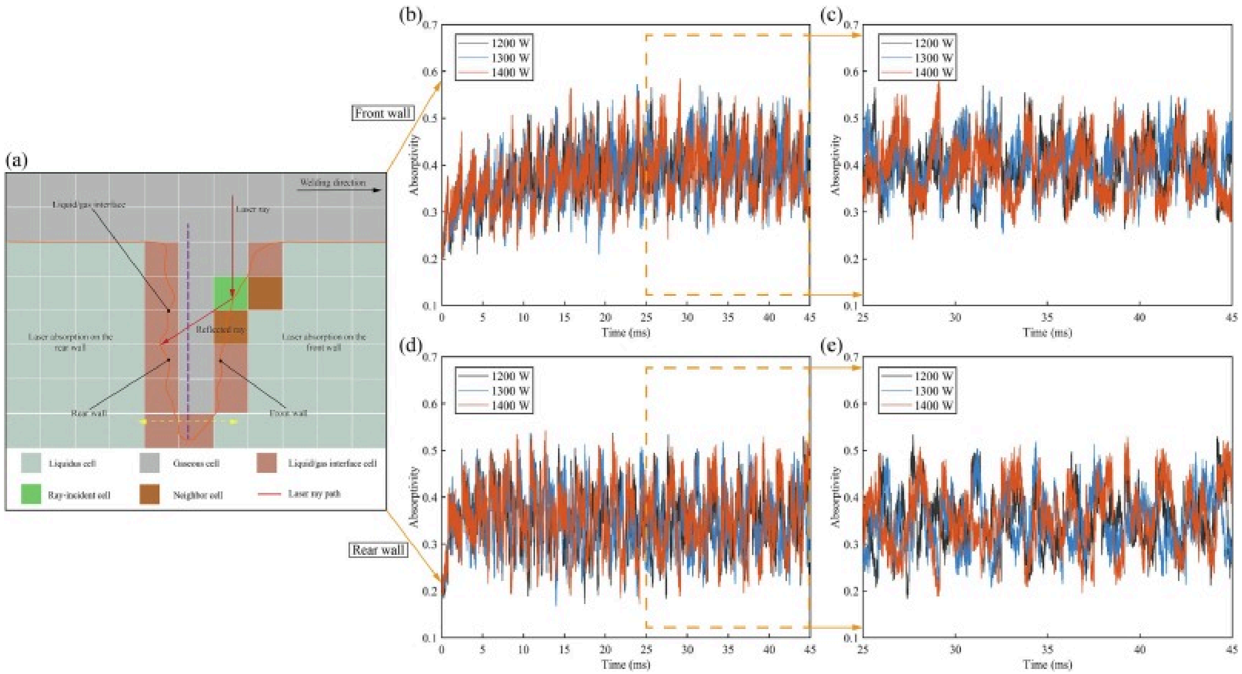

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Fig. 16. Quantitative dynamic data of laser energy absorption: (a) Schematic diagram of laser absorptivity statistics; (b) Time-resolved laser absorptivity on the front wall; (c) Time-dependent variation of laser absorptivity on the front wall from 25 ms to 45 ms; (d) Time-resolved laser absorptivity on the rear wall; (e) Time-dependent variation of laser absorptivity on the rear wall from 25 ms to 45 ms.

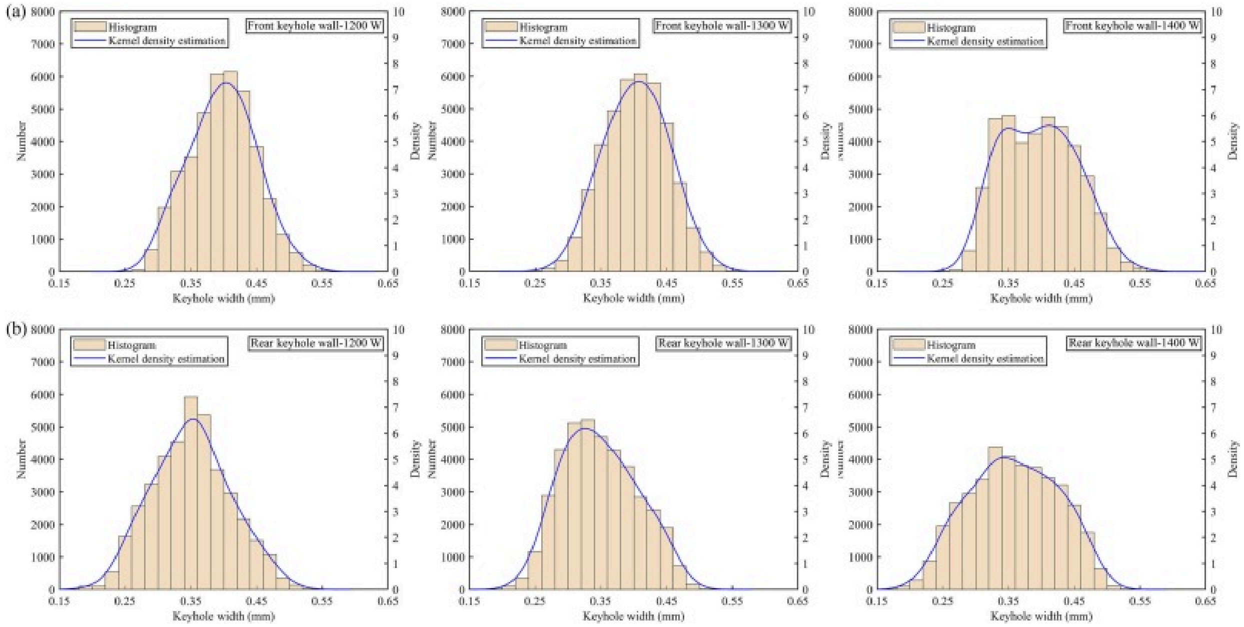

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Fig. 17. Histogram and KDE curve of the absorptivity for: (a) the front wall; (b) the rear wall.

Table 4. Laser energy absorption for simulation cases.

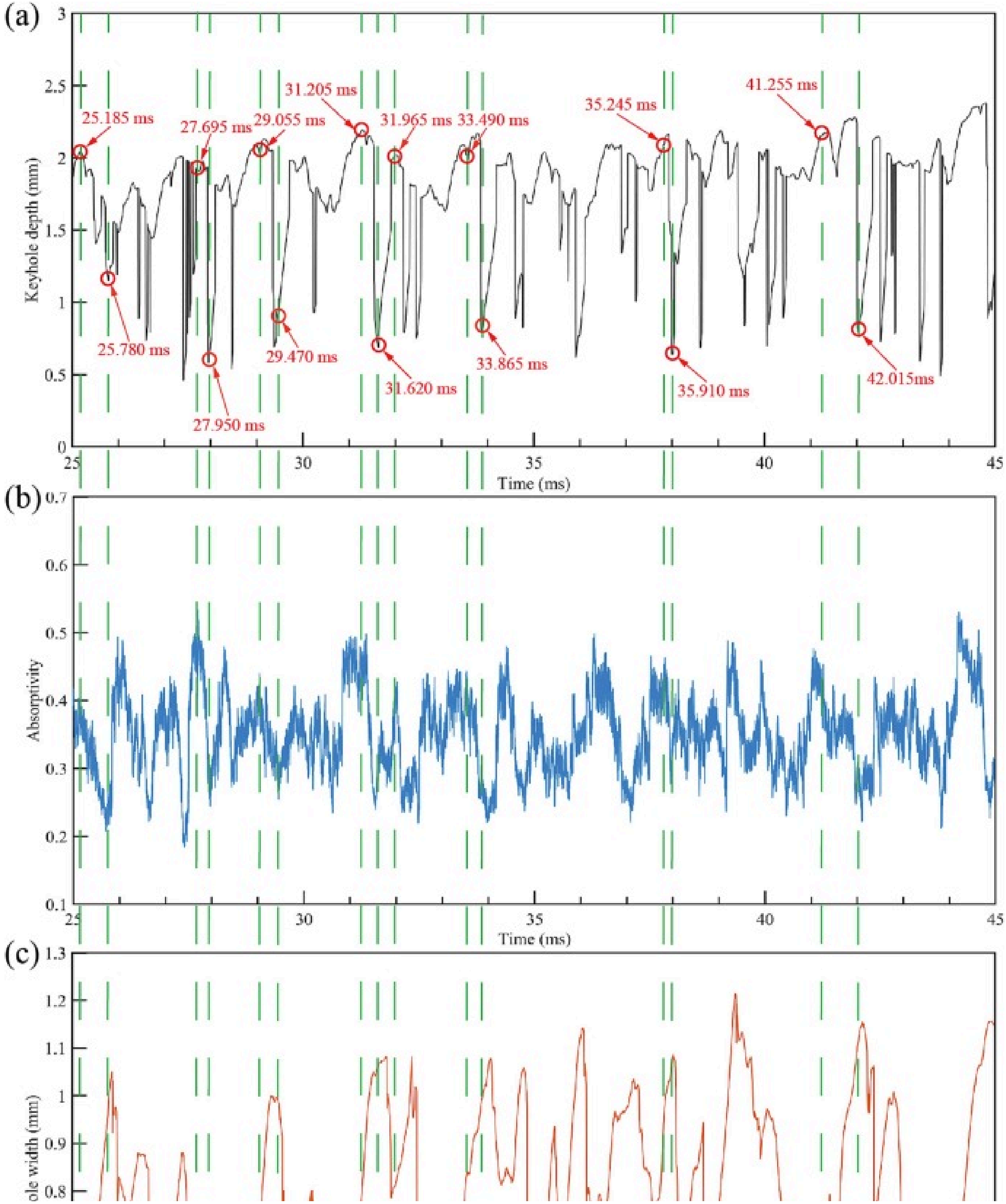
Laser absorptivity		**Case A**	**Case B**	**Case C**
Front keyhole wall	Equilibrium value (%)	39.78	40.36	39.60
	Standard deviation (%)	4.98	4.78	5.64
Rear keyhole wall	Equilibrium value (%)	35.12	34.84	35.75
	Standard deviation (%)	5.88	5.67	6.69
Entire keyhole	Equilibrium value (%)	75.11	75.33	75.47

Laser absorptivity		Case A	Case B	Case C
	Standard deviation (%)	3.40	4.21	3.84

4.4. Interrelation mechanism between laser energy absorption and keyhole dynamics

The dynamic behavior of the keyhole plays a crucial role in modulating the interaction between the incident laser beam and keyhole wall. As the keyhole fluctuates, it alters the path and number of internal reflections of the laser rays, thereby dynamically changing the local absorptivity. In 4.2 Evaluation and analysis of keyhole dynamics, 4.3 Evaluation and analysis of laser energy absorption for dynamic keyhole, the fluctuating behaviors are observed for the keyhole morphology parameters and laser absorptivity around their respective equilibrium states, suggesting strong coupling between keyhole dynamics and energy absorption. Therefore, this section utilizes the developed 3D multi-physics model to investigate the coupling mechanisms between keyhole dynamics and laser energy absorption, aiming to elucidate the underlying physical mechanism governing these fluctuating phenomena.

To understand this coupling mechanism between the keyhole and laser beam, the synchronous analysis of the keyhole morphology parameters and laser absorptivity is conducted using simulation results from Case A, as shown in Fig. 18. The numerical results indicates that the periodic variations in the keyhole depth is generally synchronized with the periodic fluctuations in the laser absorptivity. A clear periodic correlation between the keyhole depth and laser absorptivity can be observed throughout the fluctuation stage. This synchronized behavior results from the strong coupling effect between the evolving keyhole and the incident laser beam. However, the fluctuations in the keyhole width do not align with the observed variations in the keyhole depth and laser absorptivity.


(a)
Keyhole depth (mm)
Time (ms)
25.185 ms
25.780 ms
27.695 ms
27.950 ms
29.055 ms
29.470 ms
31.205 ms
31.620 ms
31.965 ms
33.490 ms
33.865 ms
35.245 ms
35.910 ms
41.255 ms
42.015ms
(b)
Absorptivity
Time (ms)
(c)
ole width (mm)

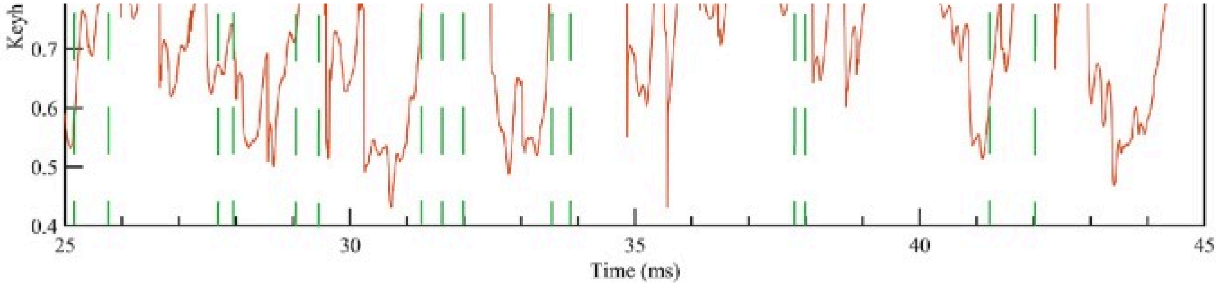


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Fig. 18. Synchronous variations in the keyhole morphology parameters and laser energy absorption (laser power: 1200 W): (a) Time-resolved keyhole depth; (b) Time-resolved laser absorptivity; (c) Time-resolved keyhole width.

The observed absorptivity behavior is intrinsically linked to the morphological features of the keyhole. To elucidate the underlying physical mechanism, the fluctuating cycle of the keyhole depth and laser absorptivity around their equilibrium states is analyzed in detail. Fig. 19 illustrates the representative snapshots of the keyhole morphology and corresponding laser energy deposition on the keyhole wall during one complete fluctuating cycle (31.205–31.965 ms). The keyhole presents the elongated morphology at 31.205 ms (Fig. 19 a2), facilitating internal beam reflections and energy deposition of the laser rays. At this point, the laser energy absorption within the keyhole reaches its peak. It is noteworthy that energy deposition on the keyhole wall is characterized by unevenness. As laser energy absorption is highly sensitive to the incidence angle and surface temperature, spatially nonuniform absorption can induce uneven vaporization, thereby generating nonuniform recoil pressure on the free surface [52]. The resulting uneven recoil pressure can in return cause perturbations on the keyhole surface and variations in the keyhole morphology over time, which subsequently leads to fluctuations in laser energy absorption. Consequently, the laser absorptivity gradually decreases as the opportunities for internal beam reflection become limited in the process of necking (Fig. 19 a2 to a3). Subsequently, when the surface tension becomes dominant, the keyhole deforms and splits into two distinct regions in shape (Fig. 18 a3), resulting in the instantaneous bubble formation. The lower part of the keyhole experiences a pronounced temperature decrease due to inadequate laser energy penetration. At 31.620 ms, a simultaneous decline in both keyhole depth and laser absorptivity to their valley values is observed (Fig. 18(a) and (b)). With the progression of the welding process, the increasing recoil pressure drives the keyhole to continuously penetrate downwards (Fig. 19 a4 to a8). During this process, the keyhole depth and laser absorptivity exhibits the general trend of synchronous increase. Finally, propelled by recoil pressure, the keyhole bottom merges with the previously generated bubble. The emergence of the extended keyhole structure indicates the beginning of a subsequent fluctuation cycle. During the entire quasi-steady process, the keyhole morphology and laser absorptivity repeatedly undergo this fluctuating pattern around their respective equilibrium states.

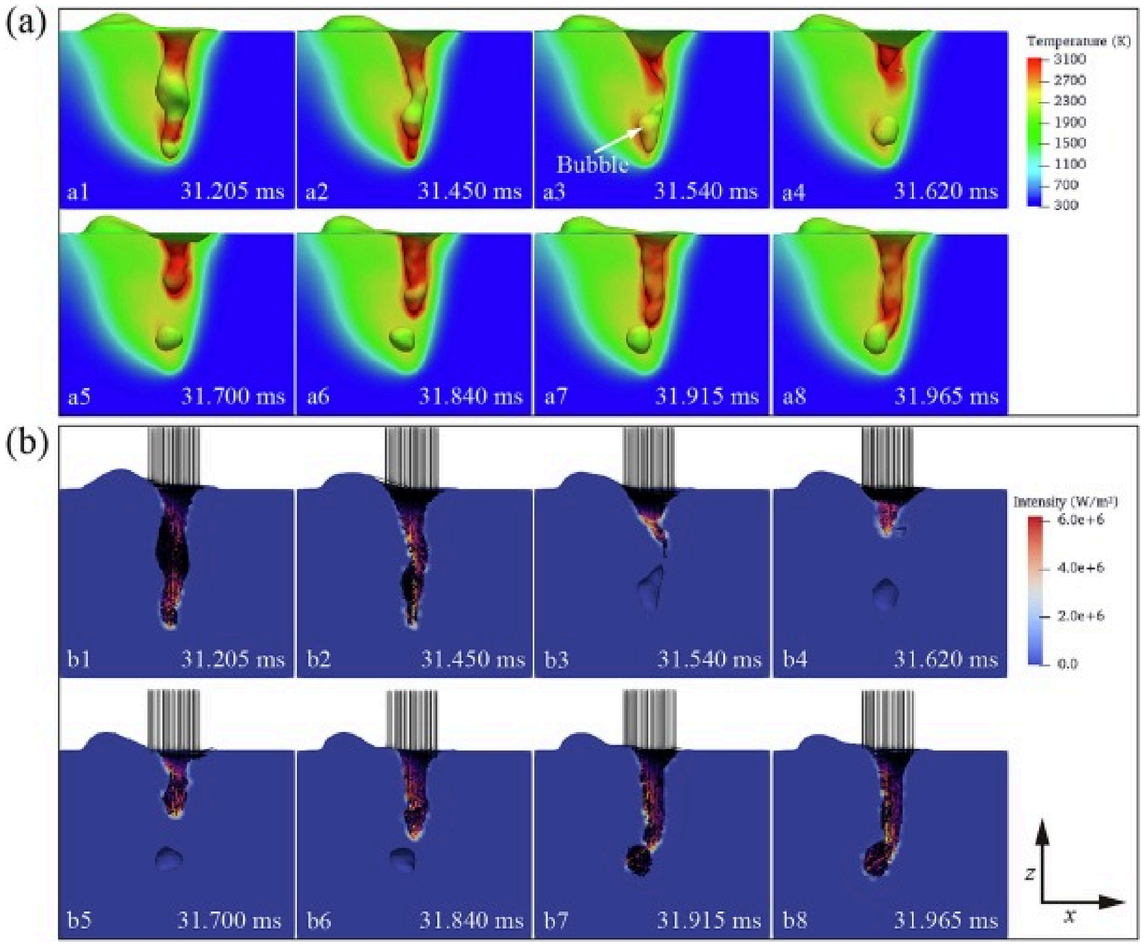

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Fig. 19. Exemplary time sequences of simulation results for the keyhole morphology and associated absorbed power distribution (laser power: 1200W).

Fig. 19(b) shows the laser propagation path and spatial distribution of laser absorption intensity. It can be observed that, in the fluctuation stage, the opened keyhole opening allows most of the incident laser rays to enter its interior without significantly affecting the initial laser energy input. Additionally, the keyhole width is predominantly influenced by the dynamics of the molten metal flow. Therefore, the variations in the keyhole width does not correlate synchronously with the temporal evolution of the laser absorptivity. Therefore, the synchronous analysis of the transient keyhole behavior and laser energy absorption reveals the underlying mechanisms of their fluctuations around the equilibrium center of the stochastic fluctuation behavior.

5. Conclusions

(1) The keyhole dynamic parameters exhibited stochastic fluctuations around the equilibrium state represented by the point of the peak probability density in the fluctuation stage. The fluctuation of the keyhole depth followed a left-skewed Gaussian distribution, where the skewed portion demonstrated significant correlation with keyhole collapse events. As a low-probability event, keyhole collapse can be regarded as a deviation from the stable process. The variation of the keyhole width approximately followed a normal distribution, and its fluctuation range and equilibrium value increased as the laser power increased.

(2) The laser absorptivity underwent periodic fluctuations around the equilibrium state point corresponding to the peak probability density in the fluctuation stage. The laser absorptivity on the entire keyhole, as well as on the front and rear walls, approximately followed Gaussian distributions. The laser absorptivity on the rear wall contributed more to the overall fluctuations in total laser absorptivity than that on the front wall.

(3) The dynamic behavior of the keyhole depth exhibited synchronous changes with the variations in the laser absorptivity. The synchronous analysis of transient results provided evidence for the physical phenomena of the keyhole behavior and laser absorptivity fluctuating around the equilibrium point revealed through statistical analysis.

CRediT authorship contribution statement

Yunfu Tian: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Changxing Li:** Visualization, Validation, Methodology, Investigation, Data curation. **Yan Xie:** Validation, Methodology, Investigation, Data curation. **Lijun Yang:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Yiming Huang:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that we have no competing financial interests and personal relationships with other people or organizations that can inappropriately influence our work.

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